KCE PATHWAYS · GRADES 4–5 · UNIT 4: SHAPES AND SPACE · WEEK 19 · MATH

The Peach Packing House — Teacher Lesson Plan & Answer Key

Learning objectives: Students find the volume of rectangular prisms by counting unit cubes and layers, use V = l × w × h and V = B × h with cubic units, find the volume of composite figures made of two prisms, find a missing dimension from a known volume, and solve packing problems that ask how many boxes fit.

Duration: 35–45 minutes
Materials: Computer or tablet with the lesson open · pencil · printed Pre & Post sheet

Lesson flow (as it appears on screen)

1. Slide 0 — Intro and name: meet Makena at the peach packing house in Fort Valley, Georgia, where Uncle Thaddeus is the packing manager, Ms. Lindqvist runs the cold rooms and Mr. Kowalski drives the truck. The student types first and last name so results are sent. Read the scoring rules together: 2 tries per question, Pathway Points only for a correct first try, and the answer plus the reason appear after the 2nd miss. Typed number answers ignore capital letters, commas in large numbers, dollar signs and a final period.
2. Slide 1 — Warm-Up, Part 1: Angles and Lines Review (4 pts, 2 each, Week 18 skill).
3. Slide 2 — Warm-Up, Part 2 (4 pts, 2 each): two typed missing-angle items.
4. Slide 3 — Pencil Break (Pre): the student opens or prints the Pre/Post sheet (grade4-5/prepost/KCE_Math_PrePost_G4-5W19.pdf) and completes Page 1 (Pre-Lesson Warm-Up) off screen before any teaching.
5. Slide 4 — Why a Packing House Runs on Volume: nonfiction connection about Georgia, the “Peach State,” and Peach County. No questions.
6. Slide 5 — Picture It: Unit Cubes and Layers. Volume is the number of unit cubes that fill a box: cubes in one layer × number of layers.
7. Slide 6 — Your Turn, Part 1: Unit Cubes and Layers (8 pts, 4 each).
8. Slide 7 — Your Turn, Part 2 (4 pts): one more problem.
9. Slide 8 — Teach: Two Volume Formulas. Uncle Thaddeus’s 12 × 10 × 6 inch boxes: V = l × w × h and V = B × h. Insist on cubic units.
10. Slide 9 — Your Turn, Part 1: Using the Formulas (8 pts, 4 each).
11. Slide 10 — Your Turn, Part 2 (4 pts): a two-step missing-dimension item.
12. Slide 11 — Teach: Composite Figures. The loading-dock steps are two prisms: find each volume, then add (never multiply the parts).
13. Slide 12 — Your Turn, Part 1: Composite Figures (8 pts, 4 each).
14. Slide 13 — Your Turn, Part 2 (4 pts): one more problem.
15. Slide 14 — Vocabulary Round-Up: 8 flip cards (volume, unit cube, cubic unit, rectangular prism, base (B), height, layer, composite figure).
16. Slide 15 — Your Turn: Using the Words (6 pts, 3 each): one multiple-choice and one typed vocabulary item.
17. Slide 16 — Teach: Packing Problems and Missing Measurements. Mr. Kowalski’s trailer: find a missing length, a number of boxes, or the space left.
18. Slide 17 — Your Turn: Thinking With Volume (6 pts, 3 each).

19. Slide 18 — Real World: The Cold Room (4 pts): a different cold-room, trailer or pallet situation in each set.
20. Slide 19 — Real World: Bins and Beds (4 pts): raised beds, bins or two cold rooms; multi-step volume.
21. Slide 20 — Think It Through: Explain and Find the Mistake (10 pts, 5 each): a “which explanation is correct?” item and a “find the mistake” item. Ask the student to say why each wrong choice is wrong.
22. Slide 21 — Fix the Sentence, Part 1 (4 pts, 2 each): typed sentences checked exactly (capital letters, quotation marks, commas, end marks, and one grammar fix). Straight or curly quotes and extra spaces are OK. The corrected sentence appears after the 2nd miss.
23. Slide 22 — Fix the Sentence, Part 2 (4 pts, 2 each): two more sentences, same rules. Where shown, both serial-comma and no-serial-comma forms of a list are accepted, and one item also accepts either verb tense.
24. Slide 23 — Match-Up Round (10 pts): 5 prisms (l × w × h, or B and h) matched to volumes, 2 pts each on the first try.
25. Slide 24 — Speed Trail (bonus XP only): 3 quick items against the student’s own best time.
26. Slide 25 — Looking Back reflection (bonus XP only).
27. Slide 26 — Boss Battle: Loading the Truck (8 pts, 4 each): two multi-step packing questions about the trailer, a platform, or a shipping box.
28. Slide 27 — Pencil Break (Post): the student completes Page 2 (Post-Lesson Challenge) of the same sheet (grade4-5/prepost/KCE_Math_PrePost_G4-5W19.pdf), plus the ★ Challenge if time allows, before seeing the score.
29. Slide 28 — Results (Packing Day Complete!): the score out of 100 is sent to KingCretot Experience. 85 or higher shows “Fort Valley Volume Master!” with the congratulations message “You packed the whole peach harvest in Fort Valley!” and a big confetti burst; below 85 shows “Good Work at the Packing House!” with encouragement (“You are getting closer every time.”), and the next attempt brings a fresh question set (sets rotate 1, 2, 3). Both messages remind the student to finish the Post-Lesson Challenge sheet.

Lesson answer key with reasoning

Every scored item the student sees, in order, across all three rotating question sets. Each item allows 2 tries and earns Pathway Points only on a correct first try. The "why" is the same reasoning shown on screen after a second missed try.

WARM-UP: ANGLES AND LINES REVIEW (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
r1_s0	Set 1: Angle addition (Week 18): angle ABD is 35° and angle DBC is 48°. Ray BD is between them. What is the measure of angle ABC? Angle ABC (degrees):	**83** (also accepted: 83°, 83 degrees, 83 degree, 83 deg)	The two angles sit side by side, so add: 35° + 48° = 83°.
r2_s0	Set 1: Lines (Week 18): two hallways in the packing house cross each other and form four right angles. The hallways are ___. [Perpendicular / Parallel / Rays / Acute]	**Perpendicular**	Perpendicular lines cross to form four right angles (90° each). Parallel lines never cross at all.
r3_s0	Set 1: Missing angle (Week 18): two angles together form a straight line. One angle is 112°. What is the other angle? Degrees:	**68** (also accepted: 68°, 68 degrees, 68 degree, 68 deg)	A straight angle is 180°. 180° − 112° = 68°.
r4_s0	Set 1: Missing angle (Week 18): a right angle is split into two angles. One of them is 29°. What is the other one? Degrees:	**61** (also accepted: 61°, 61 degrees, 61 degree, 61 deg)	A right angle is 90°. 90° − 29° = 61°.

ID	Question	Answer	Why (explain to student)
r1_s1	Set 2: Angle addition (Week 18): angle PQS is 64° and angle SQR is 59°. Ray QS is between them. What is the measure of angle PQR? Angle PQR (degrees):	**123** (also accepted: 123°, 123 degrees, 123 degree, 123 deg)	The two angles sit side by side, so add: 64° + 59° = 123°.
r2_s1	Set 2: Figures (Week 18): which figure has exactly one endpoint and goes on forever in one direction? [A ray / A line / A line segment / A point]	**A ray**	A ray starts at one endpoint and goes on forever in one direction. A line goes on forever in both directions, and a line segment has two endpoints.
r3_s1	Set 2: Missing angle (Week 18): two angles together form a straight line. One angle is 47°. What is the other angle? Degrees:	**133** (also accepted: 133°, 133 degrees, 133 degree, 133 deg)	A straight angle is 180°. 180° − 47° = 133°.
r4_s1	Set 2: Missing angle (Week 18): a right angle is split into two angles. One of them is 38°. What is the other one? Degrees:	**52** (also accepted: 52°, 52 degrees, 52 degree, 52 deg)	A right angle is 90°. 90° − 38° = 52°.
r1_s2	Set 3: Angle addition (Week 18): angle JKM is 72° and angle MKL is 26°. Ray KM is between them. What is the measure of angle JKL? Angle JKL (degrees):	**98** (also accepted: 98°, 98 degrees, 98 degree, 98 deg)	The two angles sit side by side, so add: 72° + 26° = 98°.
r2_s2	Set 3: Kinds of angles (Week 18): an angle measures 135°. What kind of angle is it? [Obtuse / Acute / Right / Straight]	**Obtuse**	135° is more than 90° (a right angle) but less than 180° (a straight angle), so it is obtuse.
r3_s2	Set 3: Two steps (Week 18): three angles together form a straight line. Two of them are 40° and 85°. What is the third angle? Degrees:	**55** (also accepted: 55°, 55 degrees, 55 degree, 55 deg)	Three angles on a straight line add to 180°. 40° + 85° = 125°, and 180° − 125° = 55°.
r4_s2	Set 3: Two steps (Week 18): three angles meet at a point and fill the whole turn around it. Two of them are 150° and 120°. What is the third angle? Degrees:	**90** (also accepted: 90°, 90 degrees, 90 degree, 90 deg)	A full turn around a point is 360°. 150° + 120° = 270°, and 360° − 270° = 90°.

YOUR TURN: UNIT CUBES AND LAYERS (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
l1_s0	Set 1: Makena packs a gift box with 1-inch cubes of peach candy. The box is 5 cubes long and 4 cubes wide, and it holds 3 layers. What is its volume in cubic inches? Cubic inches:	**60** (also accepted: 60 cubic inches, 60 cubic inch, 60 cubic in, 60 cu in, 60 in3, 60 in^3, 60 inches cubed, 60 cubic units)	Bottom layer: 5 × 4 = 20 cubes. Three layers: 20 × 3 = 60 cubes, so 60 cubic inches.
l2_s0	Set 1: The bottom layer of a box holds 18 unit cubes. The box is 4 layers tall. What is the volume? [72 cubic units / 22 cubic units / 36 cubic units / 18 cubic units]	**72 cubic units**	Each layer has 18 cubes, and there are 4 layers: 18 × 4 = 72 cubic units. (22 comes from adding 18 + 4 instead of multiplying.)
l3_s0	Set 1: Use the layer table for a box with the same number of cubes in every layer. How many cubes fill 6 layers?LayersCubes 114 228 342 6? Cubes in 6 layers:	**84** (also accepted: 84 cubic units, 84 cubic unit, 84 cubes, 84 cube, 84 unit cubes, 84 units3, 84 $units^3$)	The table adds 14 cubes for every layer, so multiply by 14: 6 × 14 = 84 cubes.

ID	Question	Answer	Why (explain to student)
l1_s1	Set 2: Makena packs a gift box with 1-inch cubes of peach candy. The box is 6 cubes long and 3 cubes wide, and it holds 4 layers. What is its volume in cubic inches? Cubic inches:	**72** (also accepted: 72 cubic inches, 72 cubic inch, 72 cubic in, 72 cu in, 72 in3, 72 in^3, 72 inches cubed, 72 cubic units)	Bottom layer: 6 × 3 = 18 cubes. Four layers: 18 × 4 = 72 cubes, so 72 cubic inches.
l2_s1	Set 2: The bottom layer of a box holds 15 unit cubes. The box is 5 layers tall. What is the volume? [75 cubic units / 20 cubic units / 30 cubic units / 150 cubic units]	**75 cubic units**	Each layer has 15 cubes, and there are 5 layers: 15 × 5 = 75 cubic units. (20 comes from adding 15 + 5 instead of multiplying.)
l3_s1	Set 2: Use the layer table for a box with the same number of cubes in every layer. How many cubes fill 7 layers?LayersCubes 116 232 464 7? Cubes in 7 layers:	**112** (also accepted: 112 cubic units, 112 cubic unit, 112 cubes, 112 cube, 112 unit cubes, 112 units3, 112 $units^3$)	The table adds 16 cubes for every layer, so multiply by 16: 7 × 16 = 112 cubes.
l1_s2	Set 3: Makena packs a gift box with 1-inch cubes of peach candy. The box is 7 cubes long and 2 cubes wide, and it holds 5 layers. What is its volume in cubic inches? Cubic inches:	**70** (also accepted: 70 cubic inches, 70 cubic inch, 70 cubic in, 70 cu in, 70 in3, 70 in^3, 70 inches cubed, 70 cubic units)	Bottom layer: 7 × 2 = 14 cubes. Five layers: 14 × 5 = 70 cubes, so 70 cubic inches.
l2_s2	Set 3: The bottom layer of a box holds 21 unit cubes. The box is 4 layers tall. What is the volume? [84 cubic units / 25 cubic units / 42 cubic units / 105 cubic units]	**84 cubic units**	Each layer has 21 cubes, and there are 4 layers: 21 × 4 = 84 cubic units. (25 comes from adding 21 + 4 instead of multiplying.)
l3_s2	Set 3: Use the layer table for a box with the same number of cubes in every layer. How many cubes fill 8 layers?LayersCubes 120 360 5100 8? Cubes in 8 layers:	**160** (also accepted: 160 cubic units, 160 cubic unit, 160 cubes, 160 cube, 160 unit cubes, 160 units3, 160 $units^3$)	The table adds 20 cubes for every layer, so multiply by 20: 8 × 20 = 160 cubes.

YOUR TURN: USING THE FORMULAS (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
k1_s0	Set 1: A wooden peach bin is 8 feet long, 5 feet wide, and 4 feet tall. What is its volume in cubic feet? Cubic feet:	**160** (also accepted: 160 cubic feet, 160 cubic foot, 160 cubic ft, 160 cu ft, 160 ft3, 160 ft^3, 160 feet cubed, 160 cubic units)	V = l × w × h = 8 × 5 × 4. 8 × 5 = 40, and 40 × 4 = 160 cubic feet.
k2_s0	Set 1: The base of a box has an area of 48 square inches. The box is 9 inches tall. What is its volume? [432 cubic inches / 57 cubic inches / 384 cubic inches / 4,320 cubic inches]	**432 cubic inches**	V = B × h = 48 × 9 = 432 cubic inches. (57 comes from adding 48 + 9.)
k3_s0	Set 1: Two steps: a box of peach jam jars has a volume of 360 cubic centimeters. It is 10 centimeters long and 6 centimeters wide. How tall is it? Height (centimeters):	**6** (also accepted: 6 centimeters, 6 centimeter, 6 cm)	Area of the base: 10 × 6 = 60 square centimeters. Height: 360 ÷ 60 = 6 centimeters. Check: 10 × 6 × 6 = 360.
k1_s1	Set 2: A cold storage room is 9 meters long, 4 meters wide, and 3 meters tall. What is its volume in cubic meters? Cubic meters:	**108** (also accepted: 108 cubic meters, 108 cubic meter, 108 cubic m, 108 m3, 108 m^3, 108 meters cubed, 108 cubic units)	V = l × w × h = 9 × 4 × 3. 9 × 4 = 36, and 36 × 3 = 108 cubic meters.

ID	Question	Answer	Why (explain to student)
k2_s1	Set 2: The base of a storage bin has an area of 35 square feet. The bin is 8 feet tall. What is its volume? [280 cubic feet / 43 cubic feet / 245 cubic feet / 2,800 cubic feet]	**280 cubic feet**	V = B × h = 35 × 8 = 280 cubic feet. (43 comes from adding 35 + 8.)
k3_s1	Set 2: Two steps: a peach box has a volume of 480 cubic inches. It is 12 inches long and 8 inches wide. How tall is it? Height (inches):	**5** (also accepted: 5 inches, 5 inch, 5 in)	Area of the base: 12 × 8 = 96 square inches. Height: 480 ÷ 96 = 5 inches. Check: 12 × 8 × 5 = 480.
k1_s2	Set 3: A wash tank for peaches is 7 feet long, 6 feet wide, and 5 feet tall. What is its volume in cubic feet? Cubic feet:	**210** (also accepted: 210 cubic feet, 210 cubic foot, 210 cubic ft, 210 cu ft, 210 ft3, 210 ft³, 210 feet cubed, 210 cubic units)	V = l × w × h = 7 × 6 × 5. 7 × 6 = 42, and 42 × 5 = 210 cubic feet.
k2_s2	Set 3: The base of a box has an area of 64 square centimeters. The box is 7 centimeters tall. What is its volume? [448 cubic centimeters / 71 cubic centimeters / 384 cubic centimeters / 4,480 cubic centimeters]	**448 cubic centimeters**	V = B × h = 64 × 7 = 448 cubic centimeters. (71 comes from adding 64 + 7.)
k3_s2	Set 3: Two steps: a storage shed has a volume of 504 cubic feet. It is 9 feet long and 7 feet wide. How tall is it? Height (feet):	**8** (also accepted: 8 feet, 8 foot, 8 ft)	Area of the base: 9 × 7 = 63 square feet. Height: 504 ÷ 63 = 8 feet. Check: 9 × 7 × 8 = 504.

YOUR TURN: COMPOSITE FIGURES (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
e1_s0	Set 1: A cooling building is made of two prisms. What is its total volume in cubic meters?Prism A: 6 meters long, 4 meters wide, 3 meters tall Prism B: 5 meters long, 2 meters wide, 2 meters tall Cubic meters:	**92** (also accepted: 92 cubic meters, 92 cubic meter, 92 cubic m, 92 m3, 92 m³, 92 meters cubed, 92 cubic units)	Prism A: 6 × 4 × 3 = 72. Prism B: 5 × 2 × 2 = 20. Total: 72 + 20 = 92 cubic meters.
e2_s0	Set 1: A sorting table is made of two prisms. Prism A is 10 feet by 3 feet by 2 feet. Prism B is 4 feet by 3 feet by 5 feet. What is the total volume? [120 cubic feet / 60 cubic feet / 3,600 cubic feet / 27 cubic feet]	**120 cubic feet**	Prism A: 10 × 3 × 2 = 60. Prism B: 4 × 3 × 5 = 60. Add: 60 + 60 = 120 cubic feet. (3,600 comes from multiplying the two volumes, and 27 comes from adding all six numbers.)
e3_s0	Set 1: Two steps: a composite storage shed has a total volume of 230 cubic feet. Prism A is 5 feet long, 4 feet wide, and 6 feet tall. What is the volume of Prism B? Cubic feet:	**110** (also accepted: 110 cubic feet, 110 cubic foot, 110 cubic ft, 110 cu ft, 110 ft3, 110 ft³, 110 feet cubed, 110 cubic units)	Prism A: 5 × 4 × 6 = 120 cubic feet. Prism B: 230 − 120 = 110 cubic feet.
e1_s1	Set 2: A cooling building is made of two prisms. What is its total volume in cubic meters?Prism A: 7 meters long, 5 meters wide, 2 meters tall Prism B: 3 meters long, 5 meters wide, 4 meters tall Cubic meters:	**130** (also accepted: 130 cubic meters, 130 cubic meter, 130 cubic m, 130 m3, 130 m³, 130 meters cubed, 130 cubic units)	Prism A: 7 × 5 × 2 = 70. Prism B: 3 × 5 × 4 = 60. Total: 70 + 60 = 130 cubic meters.
e2_s1	Set 2: A sorting table is made of two prisms. Prism A is 8 feet by 2 feet by 3 feet. Prism B is 6 feet by 2 feet by 4 feet. What is the total volume? [96 cubic feet / 48 cubic feet / 2,304 cubic feet / 25 cubic feet]	**96 cubic feet**	Prism A: 8 × 2 × 3 = 48. Prism B: 6 × 2 × 4 = 48. Add: 48 + 48 = 96 cubic feet. (2,304 comes from multiplying the two volumes, and 25 comes from adding all six numbers.)

ID	Question	Answer	Why (explain to student)
e3_s1	Set 2: Two steps: a composite storage shed has a total volume of 260 cubic feet. Prism A is 6 feet long, 5 feet wide, and 4 feet tall. What is the volume of Prism B? Cubic feet:	**140** (also accepted: 140 cubic feet, 140 cubic foot, 140 cubic ft, 140 cu ft, 140 ft3, 140 ft³, 140 feet cubed, 140 cubic units)	Prism A: 6 × 5 × 4 = 120 cubic feet. Prism B: 260 − 120 = 140 cubic feet.
e1_s2	Set 3: A cooling building is made of two prisms. What is its total volume in cubic meters?Prism A: 9 meters long, 4 meters wide, 2 meters tall Prism B: 4 meters long, 4 meters wide, 3 meters tall Cubic meters:	**120** (also accepted: 120 cubic meters, 120 cubic meter, 120 cubic m, 120 m3, 120 m³, 120 meters cubed, 120 cubic units)	Prism A: 9 × 4 × 2 = 72. Prism B: 4 × 4 × 3 = 48. Total: 72 + 48 = 120 cubic meters.
e2_s2	Set 3: A sorting table is made of two prisms. Prism A is 5 feet by 5 feet by 2 feet. Prism B is 5 feet by 3 feet by 6 feet. What is the total volume? [140 cubic feet / 90 cubic feet / 4,500 cubic feet / 26 cubic feet]	**140 cubic feet**	Prism A: 5 × 5 × 2 = 50. Prism B: 5 × 3 × 6 = 90. Add: 50 + 90 = 140 cubic feet. (4,500 comes from multiplying the two volumes, and 26 comes from adding all six numbers.)
e3_s2	Set 3: Two steps: a composite storage shed has a total volume of 300 cubic feet. Prism A is 8 feet long, 5 feet wide, and 5 feet tall. What is the volume of Prism B? Cubic feet:	**100** (also accepted: 100 cubic feet, 100 cubic foot, 100 cubic ft, 100 cu ft, 100 ft3, 100 ft³, 100 feet cubed, 100 cubic units)	Prism A: 8 × 5 × 5 = 200 cubic feet. Prism B: 300 − 200 = 100 cubic feet.

YOUR TURN: USING THE WORDS (3 POINTS EACH)

ID	Question	Answer	Why (explain to student)
v1_s0	Set 1: From the lesson: “Every job in the packing house asks the same question: how much space is inside?” The space inside a solid figure is called its ___. [Volume / Area / Perimeter / Base]	**Volume**	Volume is the number of unit cubes that fill a solid with no gaps or overlaps. Area covers a flat surface, and perimeter is the distance around it.
v2_s0	Set 1: Type your answer: “Boxes, bins, cold rooms, and truck trailers are all shaped like rectangular ___.” (one word)	**prism** (also accepted: rectangular prism, prisms)	A solid figure with six rectangle faces, like a box, is a rectangular prism.
v1_s1	Set 2: From the lesson: “Its area is B = 12 × 10 = 120 square inches.” In the formula V = B × h, the letter B stands for ___. [The area of the base / The length of the box / The number of boxes / The height of the box]	**The area of the base**	In V = B × h, B is the area of the base, the bottom face. It tells how many unit cubes fit in the bottom layer.
v2_s1	Set 2: Type your answer: a cube that is 1 unit long, 1 unit wide, and 1 unit tall is called a unit ___.	**cube** (also accepted: unit cube, cubes)	A cube that is 1 unit on every edge is a unit cube. Its volume is 1 cubic unit.
v1_s2	Set 3: From the lesson: “The steps are not one box.” A solid made of two rectangular prisms joined together is called a ___. [Composite figure / Unit cube / Right angle / Square unit]	**Composite figure**	A composite figure is made of two or more prisms joined together. You find its volume by adding the volumes of the parts.
v2_s2	Set 3: Type your answer: volume is measured in ___ units, like cubic feet or cubic inches.	**cubic** (also accepted: cubic units, cubic unit)	Volume is measured in cubic units, such as cubic inches, cubic feet, and cubic meters.

YOUR TURN: THINKING WITH VOLUME (3 POINTS EACH)

ID	Question	Answer	Why (explain to student)
t1_s0	Set 1: A cube-shaped compost bin has edges that are each 7 feet long. What is its volume in cubic feet? Cubic feet:	**343** (also accepted: 343 cubic feet, 343 cubic foot, 343 cubic ft, 343 cu ft, 343 ft3, 343 ft³, 343 feet cubed, 343 cubic units)	Every edge of a cube is the same: 7 × 7 × 7. 7 × 7 = 49, and 49 × 7 = 343 cubic feet.
t2_s0	Set 1: A box is 6 inches long, 4 inches wide, and 5 inches tall. Uncle Thaddeus doubles the height and keeps the rest the same. What is the new volume? [240 cubic inches / 480 cubic inches / 130 cubic inches / 125 cubic inches]	**240 cubic inches**	The box starts at 6 × 4 × 5 = 120 cubic inches. Doubling the height gives 6 × 4 × 10 = 240 cubic inches, which is double. (130 comes from adding 120 + 10.)
t1_s1	Set 2: A cube-shaped box of peach candy has edges that are each 9 centimeters long. What is its volume in cubic centimeters? Cubic centimeters:	**729** (also accepted: 729 cubic centimeters, 729 cubic centimeter, 729 cubic cm, 729 cm3, 729 cm³, 729 centimeters cubed, 729 cubic units)	Every edge of a cube is the same: 9 × 9 × 9. 9 × 9 = 81, and 81 × 9 = 729 cubic centimeters.
t2_s1	Set 2: Two steps: Box A is 8 by 3 by 5 inches. Box B is 6 by 4 by 4 inches. How much more space is inside Box A than Box B? [24 cubic inches / 216 cubic inches / 2 cubic inches / 4 cubic inches]	**24 cubic inches**	Box A: 8 × 3 × 5 = 120 cubic inches. Box B: 6 × 4 × 4 = 96 cubic inches. 120 − 96 = 24 cubic inches more.
t1_s2	Set 3: A cube-shaped water tank has edges that are each 6 meters long. What is its volume in cubic meters? Cubic meters:	**216** (also accepted: 216 cubic meters, 216 cubic meter, 216 cubic m, 216 m3, 216 m³, 216 meters cubed, 216 cubic units)	Every edge of a cube is the same: 6 × 6 × 6. 6 × 6 = 36, and 36 × 6 = 216 cubic meters.
t2_s2	Set 3: Two steps: how many 2-foot cube crates fit in a space that is 10 feet long, 6 feet wide, and 4 feet tall? [30 crates / 240 crates / 120 crates / 15 crates]	**30 crates**	Fit one direction at a time: 10 ÷ 2 = 5, 6 ÷ 2 = 3, and 4 ÷ 2 = 2. 5 × 3 × 2 = 30 crates. Check: 240 cubic feet ÷ 8 cubic feet per crate = 30. (240 is the volume of the space, not the number of crates.)

REAL WORLD: THE COLD ROOM (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
w1_s0	Set 1: *The Situation Ms. Lindqvist's cold room is 20 feet long, 12 feet wide, and 10 feet tall. The cooling unit and its pipes take up 150 cubic feet of the room.* Two steps: how many cubic feet of the room are left for peach boxes? Cubic feet:	**2,250** (also accepted: 2,250 cubic feet, 2,250 cubic foot, 2,250 cubic ft, 2,250 cu ft, 2,250 ft3, 2,250 ft³, 2,250 feet cubed, 2,250 cubic units)	Room: 20 × 12 × 10 = 2,400 cubic feet. Space left: 2,400 − 150 = 2,250 cubic feet.
w1_s1	Set 2: *The Situation Mr. Kowalski's refrigerated trailer is 40 feet long, 8 feet wide, and 8 feet tall inside. This morning, peach boxes fill exactly half of the space.* Two steps: how many cubic feet of the trailer are filled with peach boxes? [1,280 cubic feet / 2,560 cubic feet / 640 cubic feet / 56 cubic feet]	**1,280 cubic feet**	Trailer: 40 × 8 × 8 = 2,560 cubic feet. Half of it: 2,560 ÷ 2 = 1,280 cubic feet. (56 comes from adding 40 + 8 + 8.)

ID	Question	Answer	Why (explain to student)
w1_s2	Set 3: *The Situation Each peach box is 2 feet long, 1 foot wide, and 1 foot tall. Makena stacks boxes on a pallet to make a solid block that is 4 feet long, 4 feet wide, and 3 feet tall.* Two steps: find the volume of the block and of one box. How many boxes are in the block? Boxes:	**24** (also accepted: 24 boxes, 24 box)	Block: 4 × 4 × 3 = 48 cubic feet. Each box: 2 × 1 × 1 = 2 cubic feet. 48 ÷ 2 = 24 boxes. (It fits exactly: 2 boxes long, 4 wide, 3 high.)

REAL WORLD: BINS AND BEDS (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
w2_s0	Set 1: *The Situation Uncle Thaddeus builds 5 raised beds for young peach trees. Each bed is 8 feet long, 3 feet wide, and 2 feet deep. Garden soil comes in bags that each hold 2 cubic feet.* Three steps: how many bags of soil does Uncle Thaddeus need to fill all 5 beds? [120 bags / 240 bags / 24 bags / 48 bags]	**120 bags**	One bed: 8 × 3 × 2 = 48 cubic feet. Five beds: 48 × 5 = 240 cubic feet. Bags: 240 ÷ 2 = 120 bags. (240 forgets the last step, and 24 forgets the 5 beds.)
w2_s1	Set 2: *The Situation The packing house has 6 peach bins. Each bin is 4 feet long, 4 feet wide, and 3 feet tall. At the end of the day, each bin is only 3/4 full.* Three steps: how many cubic feet of peaches are in all 6 bins together? [216 cubic feet / 288 cubic feet / 36 cubic feet / 72 cubic feet]	**216 cubic feet**	One bin: 4 × 4 × 3 = 48 cubic feet. 3/4 of 48 = 36 cubic feet. Six bins: 36 × 6 = 216 cubic feet. (288 forgets that the bins are only 3/4 full.)
w2_s2	Set 3: *The Situation The packing house has two cold rooms. Room A is 15 feet long, 10 feet wide, and 8 feet tall. Room B has the same length and width, but it is 10 feet tall.* Two steps: how many more cubic feet does Room B hold than Room A? Cubic feet:	**300** (also accepted: 300 cubic feet, 300 cubic foot, 300 cubic ft, 300 cu ft, 300 ft3, 300 ft³, 300 feet cubed, 300 cubic units)	Room A: 15 × 10 × 8 = 1,200 cubic feet. Room B: 15 × 10 × 10 = 1,500 cubic feet. 1,500 − 1,200 = 300 cubic feet.

THINK IT THROUGH: EXPLAIN AND FIND THE MISTAKE (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
x1_s0	Set 1: Which explanation is correct? What is the volume of a box that is 5 inches long, 4 inches wide, and 3 inches tall? [60 cubic inches, because each layer holds 5 × 4 = 20 cubes and there are 3 layers / 12 cubic inches, because 5 + 4 + 3 = 12 / 60 square inches, because 5 × 4 × 3 = 60 / 20 cubic inches, because the base is 5 × 4 = 20]	**60 cubic inches, because each layer holds 5 × 4 = 20 cubes and there are 3 layers**	The base is 5 × 4 = 20, so one layer holds 20 unit cubes. Three layers hold 20 × 3 = 60 cubic inches. Adding the edges gives a length, not a volume, and volume is measured in cubic units, not square units.
x2_s0	Set 1: Find the mistake: Matteo says a bin with a base area of 30 square feet and a height of 4 feet has a volume of 34 cubic feet. What did he do wrong? [He added. He should multiply: 30 × 4 = 120 cubic feet / He should divide: 30 ÷ 4 / He should double the base first: 60 + 4 = 64 cubic feet / Nothing. 34 cubic feet is correct]	**He added. He should multiply: 30 × 4 = 120 cubic feet**	V = B × h means multiply the base area by the height: 30 × 4 = 120 cubic feet. Matteo added instead.

ID	Question	Answer	Why (explain to student)
x1_s1	Set 2: Which explanation is correct? Which holds more: Box A, 6 by 6 by 2 units, or Box B, 4 by 4 by 4 units? [Box A, because 6 × 6 × 2 = 72 cubic units and 4 × 4 × 4 = 64 cubic units / Box B, because it is the tallest box / Box A, because 6 + 6 + 2 = 14 is more than 4 + 4 + 4 = 12 / They hold the same, because both have a volume of 64 cubic units]	**Box A, because 6 × 6 × 2 = 72 cubic units and 4 × 4 × 4 = 64 cubic units**	Box A: 6 × 6 × 2 = 72 cubic units. Box B: 4 × 4 × 4 = 64 cubic units. 72 is more than 64, so Box A holds more, even though Box B is taller.
x2_s1	Set 2: Find the mistake: Yusra finds the volume of a composite figure. Prism A is 5 × 4 × 2 = 40 cubic feet and Prism B is 3 × 4 × 2 = 24 cubic feet. She says the total is 40 × 24 = 960 cubic feet. What should she fix? [She should add the two volumes: 40 + 24 = 64 cubic feet / She should subtract: 40 − 24 = 16 cubic feet / She should add all the edges: 5 + 4 + 2 + 3 + 4 + 2 = 20 / Nothing. 960 cubic feet is correct]	**She should add the two volumes: 40 + 24 = 64 cubic feet**	For a composite figure, add the volumes of the parts: 40 + 24 = 64 cubic feet. Multiplying the two volumes does not make sense.
x1_s2	Set 3: Which explanation is correct? A box has a volume of 96 cubic centimeters. It is 6 centimeters long and 4 centimeters wide. How tall is it? [4 centimeters, because 6 × 4 = 24 and 96 ÷ 24 = 4 / 86 centimeters, because 96 − 6 − 4 = 86 / 16 centimeters, because 96 ÷ 6 = 16 / 24 centimeters, because 6 × 4 = 24]	**4 centimeters, because 6 × 4 = 24 and 96 ÷ 24 = 4**	Area of the base: 6 × 4 = 24 square centimeters. Height: 96 ÷ 24 = 4 centimeters. Check: 6 × 4 × 4 = 96.
x2_s2	Set 3: Find the mistake: Bram says a cube with 5-inch edges has a volume of 15 cubic inches, because 5 × 3 = 15. What should he fix? [Multiply length, width, and height: 5 × 5 × 5 = 125 cubic inches / Use 5 × 5 = 25 cubic inches, because a cube is a square / Use 5 × 4 = 20 cubic inches, because a cube has 4 sides / Nothing. 15 cubic inches is correct]	**Multiply length, width, and height: 5 × 5 × 5 = 125 cubic inches**	A cube has length, width, and height all equal to 5 inches, so V = 5 × 5 × 5 = 125 cubic inches. 5 × 3 adds the edge three times instead of multiplying.

FIX THE SENTENCE (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
s1_s0	Set 1: Fix this sentence: makena asked, how many cubic feet does the cold room hold	**Makena asked, "How many cubic feet does the cold room hold?"**	Capitalize the name Makena, put quotation marks around her exact words, capitalize the first word of the quote, and put the question mark inside the closing quotation mark.
s2_s0	Set 1: Fix this sentence: The packing house opened it's doors at dawn.	**The packing house opened its doors at dawn.**	Its (no apostrophe) shows ownership: the doors belong to the packing house. It's always means it is.
s3_s0	Set 1: Fix this sentence: Each box holds peaches plums and nectarines.	**Each box holds peaches, plums, and nectarines.** (also accepted: Each box holds peaches, plums and nectarines.)	Use commas to separate the items in a series: peaches, plums, and nectarines.

ID	Question	Answer	Why (explain to student)
s4_s0	Set 1: Fix this sentence: the boxes on the top shelf holds 960 cubic inches each	**The boxes on the top shelf hold 960 cubic inches each.** (also accepted: The boxes on the top shelf held 960 cubic inches each.)	Capitalize the first word. The subject is boxes (plural), so use hold, not holds. End with a period.
s1_s1	Set 2: Fix this sentence: uncle thaddeus said, every crate gets a label	**Uncle Thaddeus said, "Every crate gets a label."**	Capitalize Uncle Thaddeus (a title used with a name), put quotation marks around his exact words, capitalize the first word of the quote, and put the period inside the closing quotation mark.
s2_s1	Set 2: Fix this sentence: The workers left there gloves by the door.	**The workers left their gloves by the door.**	Their shows ownership: the gloves belong to the workers. There tells where something is.
s3_s1	Set 2: Fix this sentence: To find the volume multiply the length the width and the height.	**To find the volume, multiply the length, the width, and the height.** (also accepted: To find the volume, multiply the length, the width and the height.)	Put a comma after the opening phrase To find the volume. Then use commas to separate the items in a series: the length, the width, and the height.
s4_s1	Set 2: Fix this sentence: ms. lindqvist check the cold room every morning	**Ms. Lindqvist checks the cold room every morning.** (also accepted: Ms. Lindqvist checked the cold room every morning.)	Capitalize the name Ms. Lindqvist. One person needs the verb checks, not check. End with a period.
s1_s2	Set 3: Fix this sentence: makena said, a cube has six square faces	**Makena said, "A cube has six square faces."**	Capitalize Makena, put quotation marks around her exact words, capitalize the first word of the quote, and put the period inside the closing quotation mark.
s2_s2	Set 3: Fix this sentence: Its important to measure the height in feet.	**It's important to measure the height in feet.**	It's means it is: it is important. Use the apostrophe.
s3_s2	Set 3: Fix this sentence: The trucks drove to Macon Atlanta and Savannah.	**The trucks drove to Macon, Atlanta, and Savannah.** (also accepted: The trucks drove to Macon, Atlanta and Savannah.)	Use commas to separate the items in a series: Macon, Atlanta, and Savannah.
s4_s2	Set 3: Fix this sentence: the peaches in each box was picked this morning	**The peaches in each box were picked this morning.**	Capitalize the first word. The subject is peaches (plural), not box, so use were, not was. End with a period.

MATCH-UP ROUND (2 POINTS PER FIRST-TRY MATCH)

ID	Question	Answer	Why (explain to student)
match_s0_1	Set 1: Match 2 × 3 × 4	**24 cubic units**	2 points if matched on the first try.
match_s0_2	Set 1: Match 5 × 5 × 5	**125 cubic units**	2 points if matched on the first try.
match_s0_3	Set 1: Match 10 × 2 × 3	**60 cubic units**	2 points if matched on the first try.

ID	Question	Answer	Why (explain to student)
match_s0_4	Set 1: Match B = 20, h = 6	**120 cubic units**	2 points if matched on the first try.
match_s0_5	Set 1: Match 4 × 4 × 2	**32 cubic units**	2 points if matched on the first try.
match_s1_1	Set 2: Match 3 × 3 × 3	**27 cubic units**	2 points if matched on the first try.
match_s1_2	Set 2: Match 7 × 2 × 4	**56 cubic units**	2 points if matched on the first try.
match_s1_3	Set 2: Match B = 18, h = 5	**90 cubic units**	2 points if matched on the first try.
match_s1_4	Set 2: Match 10 × 10 × 1	**100 cubic units**	2 points if matched on the first try.
match_s1_5	Set 2: Match 4 × 3 × 6	**72 cubic units**	2 points if matched on the first try.
match_s2_1	Set 3: Match 2 × 2 × 2	**8 cubic units**	2 points if matched on the first try.
match_s2_2	Set 3: Match 9 × 3 × 2	**54 cubic units**	2 points if matched on the first try.
match_s2_3	Set 3: Match B = 25, h = 8	**200 cubic units**	2 points if matched on the first try.
match_s2_4	Set 3: Match 6 × 6 × 4	**144 cubic units**	2 points if matched on the first try.
match_s2_5	Set 3: Match 8 × 5 × 3	**120 cubic units**	2 points if matched on the first try.

BOSS BATTLE: LOADING THE TRUCK (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
b1_s0	Set 1: *Makena and Mr. Kowalski Have to Plan It All Mr. Kowalski's refrigerated trailer is 40 feet long, 8 feet wide, and 8 feet tall inside. Each pallet of peach boxes is shaped like a cube with 4-foot edges. Pallets can be stacked two high.* Two steps: how many pallets fit in the trailer? [40 pallets / 20 pallets / 64 pallets / 160 pallets]	**40 pallets**	Fit the pallets one direction at a time: 40 ÷ 4 = 10 long, 8 ÷ 4 = 2 wide, 8 ÷ 4 = 2 high. 10 × 2 × 2 = 40 pallets. Check: 2,560 cubic feet ÷ 64 cubic feet per pallet = 40.
b2_s0	Set 1: Two steps: the trailer holds 40 pallets. Each pallet has 2 layers of boxes with 16 boxes in each layer. How many peach boxes are in the full trailer? Boxes:	**1,280** (also accepted: 1,280 boxes, 1,280 box)	Each pallet has 2 layers × 16 boxes = 32 boxes. 40 pallets × 32 boxes = 1,280 boxes.
b1_s1	Set 2: *Makena and Uncle Thaddeus Have to Plan It All Uncle Thaddeus builds a two-step platform for the farm stand. The bottom step is a prism 9 feet long, 6 feet wide, and 1 foot tall. The top step sits on it and is 9 feet long, 3 feet wide, and 1 foot tall. He fills the steps with gravel, and one bag of gravel fills 3 cubic feet.* Two steps: what is the total volume of the platform? [81 cubic feet / 54 cubic feet / 1,458 cubic feet / 29 cubic feet]	**81 cubic feet**	Bottom step: 9 × 6 × 1 = 54 cubic feet. Top step: 9 × 3 × 1 = 27 cubic feet. Add: 54 + 27 = 81 cubic feet. (1,458 comes from multiplying 54 × 27.)
b2_s1	Set 2: Two steps: how many bags of gravel does Uncle Thaddeus need to fill the whole platform? Bags:	**27** (also accepted: 27 bags, 27 bag)	The platform holds 54 + 27 = 81 cubic feet. Each bag fills 3 cubic feet: 81 ÷ 3 = 27 bags.

ID	Question	Answer	Why (explain to student)
b1_s2	Set 3: *Makena and Uncle Thaddeus Have to Plan It All* *The packing house is ordering a new shipping box. It must hold exactly 1,440 cubic inches. Its base is 16 inches long and 10 inches wide.* Uncle Thaddeus asks for a longer box: double the length and keep the width and height the same. What is the volume of the new box? [2,880 cubic inches / 1,440 cubic inches / 5,760 cubic inches / 1,456 cubic inches]	**2,880 cubic inches**	Doubling one measurement doubles the volume: 1,440 × 2 = 2,880 cubic inches. Check: the new length is 32, and 32 × 10 × 9 = 2,880.
b2_s2	Set 3: Two steps: how tall must the original box be to hold exactly 1,440 cubic inches? Height (inches):	**9** (also accepted: 9 inches, 9 inch, 9 in)	Area of the base: 16 × 10 = 160 square inches. Height: 1,440 ÷ 160 = 9 inches. Check: 16 × 10 × 9 = 1,440.

Pre & Post sheet answer guide

Many items are open-ended. Grade for genuine understanding and complete sentences, not exact wording. Sheet: grade4-5/prepost/KCE_Math_PrePost_G4-5W19.pdf.

PRE-LESSON WARM-UP

#	Prompt	Model answer / grading note
1	Review: missing angles	a) 180° − 125° = 55° b) 90° − 34° = 56°
2	4 rows of 5 cubes, 2 layers	4 × 5 = 20 cubes per layer; 20 × 2 = 40 cubes.
3	Area of a 6 in by 3 in bottom	18 square inches
4	Unit for space inside a box	cubic inches
5	Draw a rectangular prism; label l, w, h	Drawing shows a box with length, width and height labeled.

POST-LESSON CHALLENGE

#	Prompt	Model answer / grading note
1	9 × 6 × 5 bin; B = 42, h = 8 box	a) 9 × 6 × 5 = 270 cubic feet b) 42 × 8 = 336 cubic inches
2	Two-prism platform	Prism A: 10 × 4 × 2 = 80 cubic feet. Prism B: 5 × 4 × 3 = 60 cubic feet. Total: 140 cubic feet.
3	2-foot cube crates in 12 × 6 × 4 feet	36 crates (6 × 3 × 2 crates along each edge; or 288 ÷ 8 = 36).
4	Wendell: cube with 4-foot edges is 16 cubic feet	He found the area of one face (4 × 4) and forgot the height. Volume = 4 × 4 × 4 = 64 cubic feet.

★ CHALLENGE

#	Prompt	Model answer / grading note
1	600 cubic inches with a 12 × 10 base; double the height	B = 12 × 10 = 120 square inches; 600 ÷ 120 = 5 inches tall. Doubling the height to 10 inches doubles the volume to 1,200 cubic inches.